
Relationships

The idea of studying things via their relationships with other things. Revisiting some of the concepts we've already met, and reframing them as types of relationship, to start getting used to the idea of relationships as something quite general.

In the last chapter we saw that objects take on very different qualities in different contexts. Now we're going to see that different contexts can be provided by looking at different types of relationship. For example, if we relate people by age we get a different context from if we relate them by wealth or power.

One example we investigated was taxi-cab distance. Distance can be viewed as a relationship between locations, and the taxi-cab relationship gives us a different context from the "crow distance" relationship. There are also other possibilities — we could take one-way streets into account, or we could use walking distance, which might be different, since cars and pedestrians often have access to different routes.

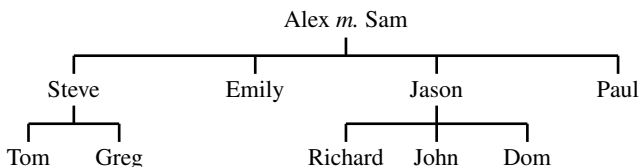
In the case of the n -hour clocks we saw a different type of relationship between numbers, in which, for example, $1 + 1$ can equal 0. This equation is really a relationship between the numbers 1 and 0. We do not have this relationship in the ordinary numbers, where we only have $1 - 1 = 0$ (and $-1 + 1 = 0$). So the existence of the relationship $1 + 1 = 0$ tells us we are in the context of the 2-hour clock, technically called the integers modulo 2.

In the case of the zero world, everything is related by being the same. Of course, everything isn't *actually* the same, but is *considered* to be the same in that world. This is an important distinction that we will keep coming back to. When we focus on context, we are looking at how things appear in *that* context. Things can appear the same in one context but not another, just like when I take my glasses off and everyone looks more or less the same to me.

In this chapter we will develop a way of dealing with relationships that heads towards the way in which category theory deals with them. We will also start drawing diagrams in the way they are drawn in category theory.

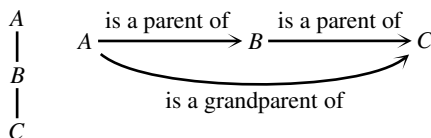
5.1 Family relationships

We sometimes depict family relationships in a family tree like this:

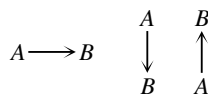


There are only three types of relationship directly depicted here: marriage, parent–child, and siblings. In fact we could view “parent–child” and “sibling” as part of the same depiction, in which case we are depicting only two types of relationship. In any case, we don’t need to depict grandparents directly, because we can deduce those relationships from two consecutive parent–child relationships.

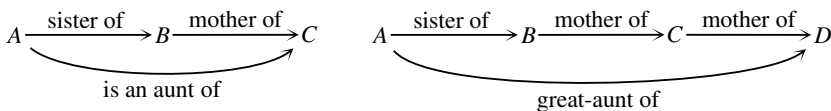
Here are two ways to depict this. The first one is a little more rigid as it depends on positions on the page.



If we represent a relationship using arrows rather than physical positioning on the page, we can draw things any of these ways up (and others) without affecting the relationship we’re expressing:



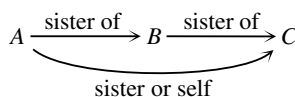
That is the point of the arrowhead. Different choices might help us visually, so the flexibility is beneficial (as flexibility typically is). The arrows also encourage us to “travel” along them to deduce other relationships such as:

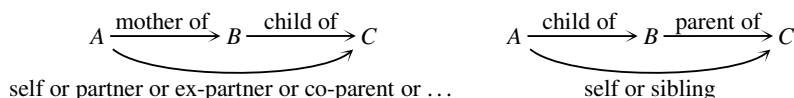


Things To Think About

T 5.1 Can you think of any situations where we can travel along two arrows and the resulting relationship could be several different things?

Note that we all have a relationship with ourself: $A \xrightarrow{\text{self}} A$ for any person A . For this among other reasons things might be ambiguous as in these examples.





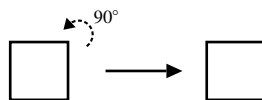
We are going to see that this way of depicting and compiling relationships is remarkably fruitful. It is general and flexible enough to be usable in a vast range of situations, and illuminating enough to have become a widespread technique in modern math and central to category theory. However we will see that we do need to impose some conditions to make sure we don't have ambiguities.

5.2 Symmetry

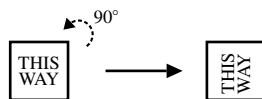
Some things might not initially seem like a type of relationship, but can be viewed like that by shifting our point of view slightly. We are going to see that categories are built from a very general type of relationship, so if we can view something as a relationship we have a chance of being able to study it with category theory. This is often how we find new examples of existing mathematical structures — it's not exactly that the example is new, but we look at it in a new way so that we see the sense in which it is an example. It's a bit like the fact that if we consider traffic like a fluid then we can understand its flow better using the math of fluid dynamics, leading to effective (and perhaps counter-intuitive) methods for easing congestion.

Symmetry is something that we might think of as a property, but we can alternatively think of it as a relationship between an object and itself. For example a square has four types of rotational symmetry: rotation by 0° , 90° , 180° or 270° . (It doesn't matter which direction we pick as long as we're consistent.)

The symmetry can be seen as this property of a square: if you rotate it by any of these angles it goes back to looking like itself.

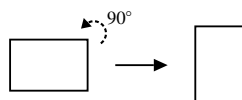


You can't tell the difference unless I put something on it.



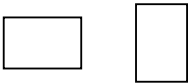
Now, the *fact* that we can do this is a property. It's a property that a rectangle, for example, doesn't possess.

A rectangle in general looks different after we rotate it 90° even without anything written on it.



In abstract math we are moving away from facts and moving towards processes. The *process* of turning a shape around is a relationship.

In the case of the rectangle it's a relationship between these two pictures.

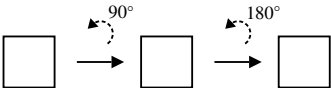


In the case of the square it's a relationship between these two pictures: the square and itself.

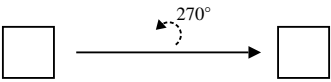


Note that now we're thinking of symmetry as a process or relationship we can ask what happens if we do one process and then another.

For example if we do these two processes



the end result is the same as doing this all in one go.



We can see this by checking that with the words written on it looks like this:



Things To Think About

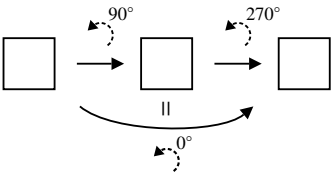
T 5.2 What happens if we do 90° and then 270° ? Remember the end result should be one of our rotations: $0^\circ, 90^\circ, 180^\circ, 270^\circ$.

If we rotate by 90° and then 270° that's the same as rotating by 360° , but this isn't in our list of rotations because it's "the same" as doing 0° .

Here "the same" means the *result* of rotating by 360° is the same as the *result* of rotating by 0° as shown here.



We could put all that information in this single diagram, which has the added benefit of looking just like the diagrams we drew for family relations previously. (The symbol looking like two short vertical lines is a rotated equals sign.)



We could depict all these relationships in a table like our previous addition and multiplication tables, to help us keep track of what is going on. So far we have these relationships.

	0	90	180	270
0				
90				
180		270		
270		0		

Things To Think About

T 5.3 You might like to try filling in the rest of this table yourself and seeing if you can see some patterns in the table. We'll come back to it in Chapter 11.

5.3 Arithmetic

We can also depict ordinary arithmetic as relationships. By “ordinary arithmetic” I mean the familiar kind of arithmetic with any familiar numbers such as whole numbers 0, 1, 2, 3, . . . which are technically called the *natural numbers*. Let us start by thinking about addition, and depict it as a relationship.

We can express addition as arrows like this.

$$1 \xrightarrow{+1} 2$$

We can then do two in succession, and see what single relationship that amounts to, as before.

$$\begin{array}{ccccc} 1 & \xrightarrow{+1} & 2 & \xrightarrow{+1} & 3 \\ & \searrow & \parallel & \nearrow & \\ & & +2 & & \end{array}$$

We could try it with a different starting point, such as this:

$$\begin{array}{ccccc} 2 & \xrightarrow{+1} & 3 & \xrightarrow{+1} & 4 \\ & \searrow & \parallel & \nearrow & \\ & & +2 & & \end{array}$$

In fact we know that $1 + 1 = 2$ in the natural numbers, so adding 1 twice will always be the same as adding 2, no matter where we start. We could depict this generally like this.

$$\begin{array}{ccccc} ? & \xrightarrow{+1} & ? & \xrightarrow{+1} & ? \\ & \searrow & \parallel & \nearrow & \\ & & +2 & & \end{array}$$

It now looks a lot like our depiction of rotations, and we can go ahead and put these in a table; it's rather large (well, infinite) but has some patterns in it that help us tell what is going to happen later without us having to fill in the whole table. Learning arithmetic is really a process of becoming familiar with these patterns, just — alas — usually without seeing the patterns drawn out.

+	0	1	2	3	4	...
0	0	1	2	3	4	...
1	1	2	3	4	5	...
2	2	3	4	5	6	...
3	3	4	5	6	...	
4	4	5	6	...		
⋮	⋮	⋮	⋮			

We started looking at the importance of patterns in Chapter 3.

5.4 Modular arithmetic

The above grid for addition of natural numbers was rather large as there is an infinite number of those numbers, but a grid for *modular* arithmetic (that is,

the n -hour clock) will be smaller. We only have a finite quantity of numbers on the n -hour clock — the numbers $1, \dots, n$ (or $0, \dots, n-1$). We regard n as being the same as 0, a bit like when we regarded 360° as being the same as 0° for rotations of a square.

Now we have different relationships. Suppose we do a 4-hour clock, so 4 is the same as 0.

We'll now get things like these:

$$1 \xrightarrow{+3} 0 \quad 2 \xrightarrow{+3} 1$$

along with combinations of arrows like this.

$$\begin{array}{c} 1 \xrightarrow{+3} 0 \xrightarrow{+2} 2 \\ \quad \quad \quad \parallel \\ \quad \quad \quad +1 \end{array}$$

As with ordinary arithmetic it doesn't matter where we start: the compiled or "composite" relation is the same.

$$\begin{array}{c} \bullet \xrightarrow{+3} \bullet \xrightarrow{+2} \bullet \\ \quad \quad \quad \parallel \\ \quad \quad \quad +1 \end{array}$$

Here the gray blobs represent places for a number to be put, like the question marks previously. In fact, as it is the processes we're interested in rather than the specific answers, we will eventually see that those gray blobs don't need to be there at all — we're really just interested in the arrows.

The arrows in this last example encapsulate the same information as simply saying $3+2 = 1$ on the 4-hour clock, and we could put all the relationships in a table as we have done for many relationships already. So far we have this:

	+0	+1	+2	+3
+0				
+1				
+2				
+3			+1	

Things To Think About

T 5.4 Try filling in the rest of the table.

1. What patterns do you see? How are they similar to the ones for rotations of a square? *Why* are they similar?
2. What symmetry can you see? What is the symmetry (visual feature) telling us about the behavior of the numbers (non-visual feature)?

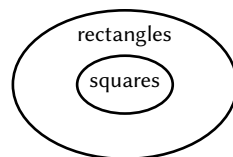
We'll come back to this in Chapter 11.

5.5 Quadrilaterals

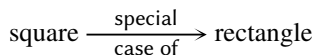
There is a meme that goes round every tax season saying "I sure am glad I learned about quadrilaterals every time quadrilateral season comes round".

This is possibly amusing, but misses the point about why we study anything: sometimes it's because the thing itself is important, but sometimes it's because it's a good arena for practicing some sort of disciplined thought process. Quadrilaterals themselves might not be particularly important for life, but I've put them here because they are a handy place to explore different types of relationship and ways of depicting them. One reason it is handy is because quadrilaterals are things we can draw and look at. Once we've got the idea, we can try this in more momentous situations or indeed anywhere we like. It's the principles that are useful, not the quadrilaterals themselves. We are going to explore "special case" relationships between types of quadrilateral.

For example, a square is a special case of a rectangle, one that happens to have all its sides of equal length. This is sometimes depicted as a Venn* diagram and this is a reasonable visualization for simple purposes.



However we are going to depict it like this:



Here are some reasons for doing this.

1. We can depict more complex interactions this way, as we'll see.
2. This abstraction looks like the way we have depicted other relationships in this chapter. In fact this is how we depict relationships in category theory and it turns out to be very fruitful for reasons including (1).

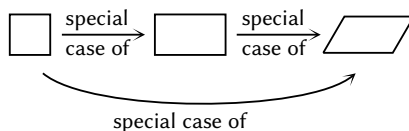
Another example of a quadrilateral relationship involves parallelograms.

Remember that parallelograms have opposite pairs of sides the same length and parallel. Thus opposite angles are the same.



A rectangle is a special case of a parallelogram as it satisfies those conditions plus more: *all* its angles are the same (right angles) not just opposite pairs.

So we have the following compilation of relationships, and this diagram of arrows that looks just like some other situations we've seen already:



The content of the "special case" diagram is that if A is a special case of B and B is a special case of C then we can immediately deduce that A is a special case of C . (One might say it is a *very* special case.) When we compile relationships in this way it is called *composition* and we say we are composing the arrows.

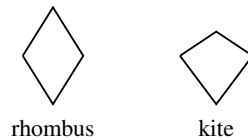
* Technically an Euler diagram. This will be the case every time I refer to a Venn* diagram.

Things To Think About

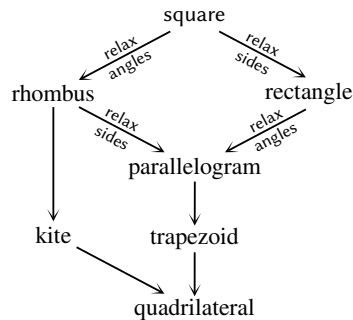
T 5.5 What other types of quadrilateral are there and what are their defining characteristics? Can you draw a diagram of all the special case relationships?

Special cases are often found by imposing extra conditions on a situation, and generalizations are found by relaxing conditions. Thinking about the conditions making a quadrilateral a square, or a trapezoid,[†] or some other special case can help us see how to relax them gradually to make more general cases.

For example a rhombus has all sides the same, and opposite angles the same, whereas a kite has adjacent pairs of edges the same, and only one pair of opposite angles the same.



For quadrilaterals the conditions involve the lengths of sides, the angles, and whether any sides are parallel. Combining all relationships between quadrilaterals we get this diagram. Note that there are two paths from “square” to “parallelogram”: it doesn’t matter in what order we relax the condition on the angles and on the sides, we get a parallelogram either way.



Things To Think About

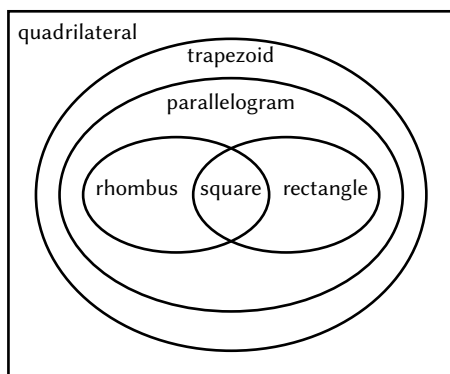
T 5.6 A trapezoid is a special case of a rhombus and of a rectangle; why don’t we need to draw those arrows? What other arrows haven’t we drawn? What would happen if we tried to draw this as a Venn* diagram?

We don’t have to draw the extra arrows as we can deduce those relationships by composing the other ones; they are the type of thing I called “very special case” earlier. On the right are a few examples.

square $\rightarrow$ parallelogram
 square $\rightarrow$ kite
 rectangle $\rightarrow$ trapezoid
 parallelogram $\rightarrow$ quadrilateral

[†] It is a point of some contention whether the following sorts of things should be allowed to count as a trapezoid (or “trapezium” in British English): I prefer to say yes as it fits better with the principle of gradually relaxing conditions. In order to disallow parallelograms from being trapezoids we would have to *add* a condition, not just relax a condition. In abstract mathematics we tend to prefer generalizing so that previous ideas are special cases rather than thrown out.

If we try to put all the types of quadrilateral into a Venn* diagram it's hard. We can get as far as the diagram below but then we have a problem.



- All rhombuses are kites, so the kite circle needs to contain the entire rhombus circle.
- A kite that is not a rhombus is also not a parallelogram or a trapezoid, so the kite circle needs to stick out into the random quadrilateral part without overlapping the parallelogram or trapezoid circles.

This means we would either have to have a disconnected “kite circle”, or have some empty intersections. Either option breaks the connection between the actual situation and the visual representation of it. The point of a visual representation is that it is supposed to clarify the relationships in question. This is a sense in which the arrow depiction has more possibilities for helping us understand more complex interactions.

In fact, in the diagram with arrows I implicitly used the height on the page as well as the arrows, as the heights correspond to how special the objects are, with “square” at the top, and things becoming progressively more general down the page. Next we will see another example where the heights on the page can be invoked to help us organize our diagrams.

5.6 Lattices of factors

We have already looked at relating numbers to each other via addition, with diagrams like this:

$$a \xrightarrow{+2} b$$

We could try this with multiplication too. This gives us a very different context. For example, if we relate numbers by addition we can build all the natural numbers (1, 2, 3, 4, ...) just starting with the number 1. But if we relate them by multiplication we can't get anywhere with the number 1; if we keep multiplying by 1 we just get 1. In fact there is no single number that we can use to get all the natural numbers by multiplication. In this section we're going to investigate the relationships between numbers by multiplication, using diagrams of arrows to help us visualize structures.

Consider the number 30. If we think about all its factors, those are the numbers a related to it as shown.

$$a \xrightarrow{\times \text{ something}} 30$$

The factors of 30 are all the numbers that “go into” 30. We can write them out in a list: 1, 2, 3, 5, 6, 10, 15, 30. This has no indication of any relationships between them; we have taken them entirely out of context.

Instead we can draw in all the relationships where anything is a factor of anything else, such as this.

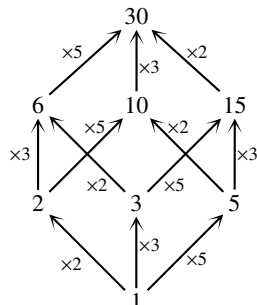
$$2 \xrightarrow{\times 3} 6$$

As usual, we don't need to draw arrows that we can deduce from composite arrows like this one.

$$2 \xrightarrow{\times 3} 6 \xrightarrow{\times 5} 30$$

$\parallel$
 $\times 15$

Here is the whole diagram. As well as omitting composite arrows I have omitted arrows showing that each number is a factor of itself. Those omissions make the diagram tidier without changing the information. In fact we could also omit the labels on the arrows, as we can deduce what each one represents by looking at the endpoints.

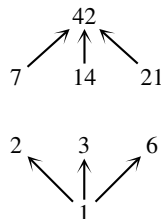


This diagram is now a cube shape, and I find that much more interesting than a list of factors in a row.

Things To Think About

- T 5.7** 1. What does each face of the cube represent? What do you notice about the labels on the edges?
2. Why did this turn out to be a cube? What other numbers will produce a cube? What shapes will other numbers produce if not cubes? You could try 42, 24, 12, 16, 60 and anything else you like.
3. How can you tell what shape will be produced before drawing it?

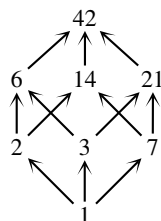
We could try this with 42. This has eight factors just like 30 does, so it has a chance of also being a cube. You might be tempted to arrange the factors in size order as in the case of 30, like this:



The top and bottom are fine but we have some problems in the middle.

- Nothing goes into 7 except 1 (and itself) as it is prime.
- We have factors on the same level: 7 goes into 14 and 21, and 2 and 3 go into 6. This didn't happen before, where all arrows went up a level.

The first point gives us a clue to how we can fix this: nothing goes into 7 except 1 (and 7) so we should move it down a level. Some things go into 6 other than 1 so we should move it up. This produces a cube diagram just as 30 did.



Things To Think About

T 5.8 What do the numbers at each level have in common with each other? Can you build on that idea to find a level of abstraction that makes this “the same” situation as the cube for 30?

The bottom number is always going to be 1 as nothing goes into it. Directly above that we'll have all the numbers (except 1) divisible by nothing except 1 and themselves — that is, the prime numbers. Note that 1 is not prime and also is not at this level; we will come back to this point.

At the next level we have the numbers that are a product of two of the primes from the level below.

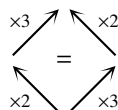
$$\begin{aligned} 6 &= 2 \times 3 \\ 14 &= 2 \times 7 \\ 21 &= 3 \times 7 \end{aligned}$$

At the top we have 42, a product of three primes.

$$42 = 2 \times 3 \times 7$$

This is the key to “why” this is a cube: 42 has three distinct prime factors.[†] In this depiction each one gives a dimension, so we have a 3-dimensional object. Notice that the parallel edges are all the same type of relationship — either $\times 2$, $\times 3$ or $\times 7$.

The square faces tell us things like the fact that multiplying by 2 and then 3 is the same as multiplying by 3 and then 2, i.e. that multiplication is commutative.

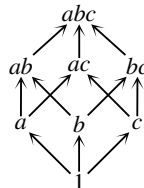


Commutativity of multiplication (together with associativity) tells us that each way of building 42 by multiplication is “the same”, and in this diagram all those different ways are represented by different paths from 1 to 42. We are

[†] We might say it has three “different” prime factors but usually in mathematics the technical word is “distinct” as it is a little more precise in broader contexts. For example we can speak of three “distinct” points in the sense that they are separate from each other, even if as points they don't look different. Differentness is subjective; it depends what criteria we use.

going to see that in category theory this is called commutativity of a diagram, whatever the diagram represents.

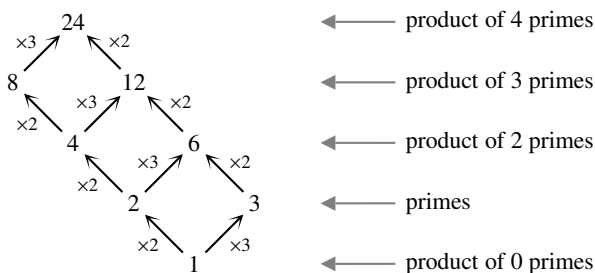
The upshot is that both 30 and 42 are a product of three distinct primes, which we could call a, b, c so that both cubes are instances of the one shown here, in which a, b , and c give us a dimension each:



I think this type of diagram vividly depicts the idea of building numbers by multiplying primes together, and gives us another way to understand why 1 is excluded from being prime: it is 0-dimensional here so we can't build any shapes from it at all.

What about a number that is not of the form $a \times b \times c$ for distinct primes a, b, c ? We will now try $24 = 2 \times 2 \times 2 \times 3$. The factors are 1, 2, 3, 4, 6, 8, 12, 24. There are eight factors so it might be tempting to try and make it a cube. But we know that a cube has three dimensions produced by three distinct prime factors, whereas here we only have two distinct prime factors, 2 and 3, with 2 being repeated several times. We are only going to put 2 in the diagram once to avoid redundancy; we are going to see that the repeatedness of the factor appears in a different, more geometric way.

The key is to remember the idea about the levels: we should put 1 at the bottom, then the primes, then the products of two primes, then three, and so on. Note that for these levels we are not thinking about *distinct* primes, so for example $4 = 2 \times 2$ and this counts as a “product of two primes”, whereas $12 = 2 \times 2 \times 3$ which is a “product of three primes”. Here is the whole diagram with the levels marked in.



Things To Think About

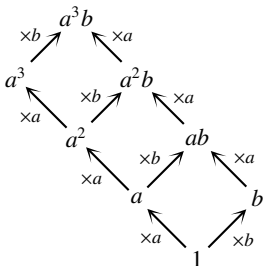
T 5.9 How many dimensions does this shape have, and why? What visual feature comes from the fact that 2 is a repeated factor? What other numbers will have the same shape? What is the abstract version of this diagram, like when we expressed the cube using a, b, c ?

This shape is 2-dimensional: it consists of three (2-dimensional) squares side by side. The two dimensions are given by $\times 2$ and $\times 3$ — they arise from the fact that 24 has only two *distinct* prime factors. The fact that 2 is a repeated factor (three times) appears visually in the fact that there are three squares in a row. This is because we can travel three times in the “ $\times 2$ ” direction.

Any other number will have the same shape if it has two distinct prime factors, one repeated three times, i.e. $a^3 \times b$, $a \neq b$, such as:

$$\begin{aligned} 3 \times 3 \times 3 \times 2 &= 54 \\ 2 \times 2 \times 2 \times 5 &= 40 \end{aligned}$$

Here the symbol $\neq$ means “not equal to”.

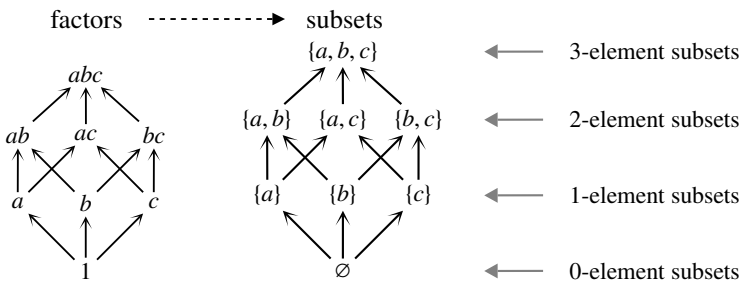


Things To Think About

T 5.10 Can you now deduce what shape will arise from *any* number, based on its prime factorization? In particular, what numbers will be 1-dimensional? Can you try drawing a 4-dimensional example, e.g. $210 = 2 \times 3 \times 5 \times 7$?

This might all seem like little more than a cute game for turning numbers into shapes, but we’re now going to see a powerful consequence of pursuing the further levels of abstraction.

Once we have reached the level of abstraction of a, b, c it doesn’t really make any difference if we multiply the numbers together or not. Consider the diagram of the abstraction of 30 and 42. It might as well be a diagram of subsets of $\{a, b, c\}$. This denotes a set of three things or “elements”, a, b, c , and a subset of this is then any set containing some combination of those elements: possibly all of them, none of them, or something in between.



The symbol $\emptyset$ denotes the empty subset, the one with no elements.

An arrow like the one on the right now shows the relationship “ A is a subset of B ”.

$$A \longrightarrow B$$

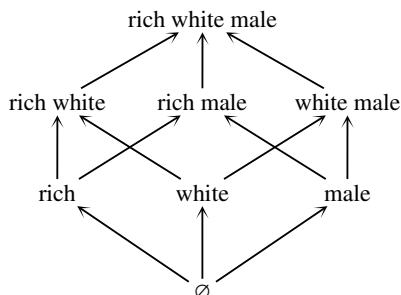
To make a tidier diagram, we only need to draw arrows for situations where exactly one element is “added” to the subset (thrown in, not actually +). When two elements are added in we can break the process down into adding one element at a time.

We can depict this in a familiar diagram.

$$\begin{array}{ccccc} \{a\} & \xrightarrow{\text{add } b} & \{a, b\} & \xrightarrow{\text{add } c} & \{a, b, c\} \\ & & & \searrow \text{add } b \text{ and } c & \\ & & & \text{||} & \end{array}$$

Now for the point of the abstraction: at this level of abstraction we can have a, b, c being *any* three things, not necessarily prime numbers, and not necessarily numbers at all because we are not trying to multiply them.

For example they could be three types of privilege, such as rich, white, male. We get this diagram:[†]



Each arrow depicts hypothetically gaining one type of privilege while everything else about you stayed the same: the theory of privilege says you would be expected to be better off in society as a result of that one change.

This depicts some important things about privilege that are often misunderstood: white privilege does not mean that *all* white people are better off than *all* non-white people, nor even that they are better off on average, nor that they are better off in every way. It is just about the abstract idea of keeping everything else fixed and moving up an arrow.

For example some white men who did not grow up rich think they do not have white privilege. However there is an arrow

non-rich non-white men $\longrightarrow$ non-rich white men

representing the gain in privilege of whiteness. This means that non-rich men who are *not* white are even worse off. However there might well be (and indeed there are) rich non-white men who are better off. But there is no arrow from rich non-white men to non-rich white men, so the theory of privilege does not say anything about the relative situations of these two groups.

An abstract explanation is that some people are confusing the hierarchy of *overall privilege* with the hierarchy of *number of types of privilege*. This is

[†] This previously appeared in *The Art of Logic*.

analogous to thinking we could make the cube of factors of 42 by putting the smallest numbers at the bottom level above 1. That didn't work because $6 < 7$ so 6 is lower in the hierarchy of sheer size, but 6 has more prime factors than 7 so is higher up in the hierarchy of number of factors.

Later we will see that these are questions of

1. totally ordered sets vs partially ordered sets, and
2. order-preserving vs non-order-preserving maps.

For now it is enough to appreciate that the formalism of drawing these diagrams of arrows can help us gain clarity about different interactions between things, and, as I've aimed to demonstrate with the examples in this chapter, that the formalism of these arrows is very flexible and can be used in a remarkably wide range of situations.

It may seem unusual to discuss issues of privilege and racism in relation to abstract mathematics. In fact I am sometimes asked if I'm not worried that talking about social justice issues in math will put some people off math, those who don't agree with me about social justice issues. The thing is that when I *didn't* talk about social justice issues in math, nobody asked me if I was worried about putting off those who *do* care about those issues. I think there are plenty of existing math books and classes that don't touch on those issues, and that talking about them includes and motivates many people who have previously not felt that math is relevant to them. I would like to stress that you don't have to agree with my stances in order to see the logical structures I am presenting in them. I think it's important for us all to be able to see the logical structures in everyone's points of view, whether we agree with them or not. I think abstract mathematics helps with that.

In the next chapter we will discuss the formalism of mathematics and why it can be helpful for progressing through the theory while also being offputting and obstructive when you are new to it.